

SHAGUN MOHTA

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EDUCATION

Indraprastha Institute of Information Technology, Delhi (IIIT Delhi)

Jan 2020 – Jun 2024

Bachelor of Technology, Computer Science (CSSS)

GPA: 8.19 / 10.0

WORK EXPERIENCE

Data Engineer, Micron Technology

Jul 2024 – Present

- Owned end to end development of **People Analytics Hub**, Micron's enterprise HR analytics platform enabling secure role based access to sensitive workforce data for **250+** HRBPs and senior leaders.
- Developed high performance user interface for executive leadership, implementing **client-side caching** and **rendering optimizations** to improve page responsiveness by **35%**.
- Automated natural language to SQL query generation by building **Snowflake Cortex** semantic models and vector search services, reducing manual query effort by **50%**
- Built a **RAG** based **HR chatbot** implementing **Agentic AI** and **REST APIs** workflows for low-latency conversational access to headcount and labor cost data.
- Built **scalable ETL pipelines and data processing workflows** to orchestrate data migration from **MSSQL, MySQL and SAP HANA** to **Snowflake**, maintained 95% uptime handling 50+ GB ingestions.
- Developed automated data testing frameworks, implementing scheduled data metric functions, improving downstream data quality by **40%** and Mean Time to Detect (MTTD) by **65%**

Data Analytics Intern, ServiceNow **Certificate**

May 2023 – Jul 2023

- Optimized legacy Snowflake stored procedures by refactoring query logic and implementing efficient partition pruning; achieved a **20.69% improvement in execution times** and reduced warehouse compute costs.
- Developed and deployed SQL data views in Snowflake supporting **8 production KPIs** used in the ServiceNow KPI dashboard.

Student Researcher, TavLab IIIT-D

Sep 2023 - Feb 2024

- Generated **3,000+** novel spike protein sequences using **state-of-the-art** transformer models to enhance **SARS-CoV-2** vaccine potential and proactively predict viral variations.
- Preprocessed **388** DNA sequences from **GISAID**, translated DNA to protein, and aligned with **MUSCLE** to standardize **1,273** amino acids.
- Computationally validated synthetic sequences for antigenicity, structural stability, and RMSD against the reference Wuhan strain using **AlphaFold**, **NetMHCpan**, **PyMol** and **DDgun**.

PROJECTS

CodeColab **Website**, **Github**

React.js, Tailwind CSS, Node.js, Express, Firebase, AWS

- Built a real-time collaborative coding platform (Skrribl inspired) with synchronized code editing, shared whiteboard and live messaging, ensuring **sub-400ms latency** for concurrent users.
- Developed an **AI-powered testing judge** using **Gemini 2.5 Flash-Lite LLM** to generate edge-case test suites and provide structured time/space complexity analysis for code submissions.
- Implemented **2,500+ LeetCode problems** across **40+ DSA topics** via **GraphQL**, deployed on **AWS** serverless stack (Lambda, API Gateway, S3).

Course Recommendation System **Github**

React.js, Bootstrap, Python, FastAPI, Sentence-BERT

- Developed a course recommendation web app for students by scraping **40+ IIITD electives**, parsing student transcripts and computing personalized suggestions through **Sentence-BERT** embeddings and **cosine similarity**.
- Exposed a **FastAPI** backend recommendation engine, achieving **sub-3s latency**, built a responsive UI with instant result rendering.

SKILLS

Languages/Frameworks Java, Python, C/C++, JavaScript, HTML/CSS, SQL, React, Angular, Node.js, Express
Tools Snowflake, MySQL, AWS, Google Cloud, Power BI, Docker, Firebase, Git, Apache NiFi

ACHIEVEMENTS

- Walmart CodeHers 2024 Finalist**, ranked among the top **0.6% (628/96,357)** female coders nationwide.
- Solved **500+** DSA problems across **LeetCode** and **GeeksforGeeks**;
- Teaching Assistant for **Introduction to Programming** and **Multivariable Calculus** courses to a cohort of **300+** B.Tech students at **IIIT Delhi**.
- Mentored primary school teachers at **Chotuu Ki Education**, NGO in Microsoft Excel, creating hands-on labs and tutorials for daily administrative tasks.